

① BCS Driven system:

$$\dot{p} = -i[H_{BCS}, p] + \gamma_0 \mathcal{D}[p]$$

$$H^{BCS}(t) = \sum_k' (c_k^\dagger, c_{-k}) \begin{pmatrix} \xi_k & -\Delta \\ -\Delta & -\xi_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} + \xi_k$$

$$= (c_k^\dagger, c_{-k}) \begin{pmatrix} \xi_k c_k + \Delta c_{-k}^\dagger \\ \Delta c_k - \xi_k c_{-k}^\dagger \end{pmatrix} =$$

$$= c_k^\dagger c_k \xi_k + \Delta c_k^\dagger c_{-k}^\dagger + \Delta c_{-k} c_k - \xi_k c_k c_{-k}^\dagger =$$

$$\rightarrow \xi_k (c_k^\dagger c_k + c_{-k} c_{-k}^\dagger) + \Delta (c_k^\dagger c_{-k}^\dagger + c_{-k} c_k) - \xi_k c_k c_{-k}^\dagger$$

Bozolyonov: $\begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} u & v \\ -v^* & u \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix}$

↳ glij notes: x opp

gap: $\Delta_k(t) = \sum_{nn'} V_{nn'} \text{Tr}[c_k^\dagger c_{k'} p(t)]$

$$\rightarrow E_k(t) = \pm \sqrt{\xi_k^2 + \Delta_k^2(t)} \quad \text{in} \quad N_k(t) = \frac{1}{2} \left(1 + \frac{\xi_k}{E_k(t)} \right), \quad N_{-k} = \frac{1}{2} \left(1 - \frac{\xi_k}{E_k(t)} \right)$$

→ simple modeliranje kopeli:

$$\mathcal{D}[\cdot] = \mathcal{D}^{(T_1)}[\cdot] + \mathcal{D}^{(T_2)}[\cdot]$$

$$\mathcal{D}^{(T)} p = \sum_{s_1, s_2} L_p L^\dagger - \frac{1}{2} \{p, L L^\dagger\}$$

$$L_{kp}^{(s_1)} = \sqrt{\gamma(T)} \alpha_k^{(s_1)} \alpha_p^{(r)}$$

$$\alpha^{(+)} = \alpha^\dagger$$

$$\alpha^{(-)} = \alpha$$

$$k = (k, \sigma)$$

$$p(t) = \frac{1}{2} \exp \left[\sum_k \mu_k(t) \alpha_k^\dagger \alpha_k \right]$$

$$\langle n_k \rangle = n_k = \langle \alpha_k^\dagger \alpha_k \rangle$$

trace

$$\rightarrow \text{EOM: } \langle \dot{n}_k \rangle = \text{Tr}[n_k \mathcal{D}_{\text{eff}}] = \gamma_0 \sum_j \langle L_j^\dagger n_k L_j \rangle - \langle L_j^\dagger L_j n_k \rangle$$

$$s_1 s_2 = \pm 1$$

$$\rightarrow \langle L_j^\dagger n_k L_j \rangle \propto \sum_{p_2} \langle \alpha_p^\dagger \alpha_z^\dagger \alpha_k^\dagger \alpha_k \alpha_p \alpha_z \rangle \quad \text{6 terms.}$$

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